

Report
Of
**"PBL NEXUS – A SMART WEB-BASED PROJECT & TASK
MANAGEMENT
SYSTEM FOR PROJECT-BASED LEARNING"**

Submitted
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B.Tech CSE



By
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CANDIDATE'S DECLARATION

I, Priyanshu Dixit, Roll Number CS-2341902, do hereby declare that the internship report titled “PBL Nexus – A Smart Web-Based Project & Task Management System for Project-Based Learning” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been duly quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

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I express my heartfelt gratitude to my parents and family for their constant encouragement and blessings throughout this journey.

Signature: _____

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INTERNSHIP COMPLETION CERTIFICATE



Certificate of Internship | Certificate ID: qveto/5aJ6dW4hx1 | Issue Date: 17-08-2026

CERTIFICATE OF TRAINING



Certificate of Training | Certificate ID: qveto/sPTnr1KE7h | Issue Date: 17-08-2026

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during a summer internship undertaken with Qveto Technologies from 07 June 2026 to 08 September 2026, in the domain of Web Development. During the course of the internship, the project “PBL Nexus” was designed and developed – a full-stack, web-based Project-Based Learning (PBL) Management System intended to streamline the process of assigning, tracking, submitting and grading academic project tasks between administrators/instructors and students. The internship provided hands-on exposure to real-world full-stack web development practices, including frontend UI/UX design, REST API development, authentication flows, file handling and cloud deployment.

4.2 Organization Profile

Qveto Technologies is a technology solutions and training organization (qveto.in) that offers internship and training programs to students in areas such as Web Development, and issues ISO 9001:2015-certified training and internship certificates upon successful completion. The organization focuses on providing project-driven, industry-oriented learning experiences that help interns build practical, deployable software products rather than purely theoretical assignments.

4.3 Problem Statement

Academic institutions running Project-Based Learning (PBL) programs typically rely on manual or fragmented methods – spreadsheets, emails and in-person follow-ups – to assign tasks to students, track their progress, collect submissions and record grades. This process is time-consuming, error-prone and offers little real-time visibility to either instructors or students. There was a clear need for a centralized, easy-to-use web platform where administrators could assign tasks and grade submissions, while students could view their tasks, update progress, upload deliverables and check their grades – all from a single system.

4.4 Project Objectives

- To design a secure, role-based login system distinguishing Admin and Student users.

- To enable administrators to add users, assign tasks with due dates, and manage the overall workflow.
- To allow students to view assigned tasks, update task status, and submit project files with notes.
- To build a grading module allowing admins to review submissions, assign marks and provide feedback.
- To provide real-time dashboards summarizing task status and submission statistics for both roles.
- To deploy the system as a live, cloud-hosted application accessible from any device.

4.5 Scope of the Project

The scope of PBL Nexus covers the complete lifecycle of a project-based academic task: user management, task assignment, submission handling (including file uploads), review and grading, and status tracking through dashboards. The system is built as a lightweight, browser-based application requiring no installation, making it usable across institutions, batches and devices. The current scope focuses on two roles – Admin and Student – and can be extended in the future to include faculty mentors, multi-file submissions, plagiarism checks and analytics.

4.6 Technologies and Tools Used

- Frontend: HTML5, CSS3 (custom design system with CSS variables, responsive layouts), and vanilla JavaScript (Fetch API, async/await, DOM manipulation).
- Typography & UI: Google Fonts (Playfair Display, DM Sans), a dark-themed “luxury academic” design language with gold accent colours.
- Backend: Node.js with a RESTful API (Express-style routes) handling authentication, user management, task management, submissions and grading.
- Database: A relational/managed database layer accessed through the backend API for storing users, tasks, submissions and grades.
- File Handling: multipart/form-data uploads (via FormData API) for student project-file submissions, served back to admins as downloadable links.

- **Deployment/Hosting:** Backend deployed on Railway (pbl-nexus-production.up.railway.app); frontend served as static pages.
- **Client-side State:** Browser localStorage used to persist the logged-in user's session and role across pages.
- **Version Control & Tools:** Git for source control, browser DevTools for debugging and testing REST endpoints.

4.7 System Architecture

PBL Nexus follows a client-server architecture. The frontend is a set of static HTML/CSS/JS pages (Login, Student Dashboard, Admin Dashboard, Add User, Assign Task, Task Submission) that communicate with a centralized backend REST API over HTTPS. The backend exposes endpoints for authentication (/login), user management (/adduser, /users), task management (/addtask, /alltasks, /tasks/:id), submissions (/submit, /submissions) and grading (/grade, /grades/:id). On successful login, the user's role (admin/student) is used to route them to the appropriate dashboard and to guard access to admin-only pages. All data operations are performed asynchronously via fetch() calls returning JSON, keeping the frontend fully decoupled from the backend implementation.

4.8 Methodology

- **Requirement Analysis:** Identified the core user roles (Admin, Student) and the key workflows – task assignment, submission, grading.
- **UI/UX Design:** Designed a consistent, dark, professional theme with reusable components (cards, badges, modals, toasts) across all pages.
- **Iterative Development:** Built and tested each module independently – authentication, dashboards, task assignment, submission upload, and grading – before integrating them.
- **API Integration:** Connected each frontend page to the live backend hosted on Railway using the Fetch API, with error handling for offline/backend-down scenarios.
- **Testing:** Manually tested each workflow end-to-end (admin adds user → assigns task → student submits → admin grades → student views grade).

- **Deployment:** Deployed the backend to Railway and verified the complete flow in a production-like environment.

4.9 Expected Outcomes

By the end of the internship, PBL Nexus was successfully delivered as a working full-stack web application with a functioning authentication system, role-based dashboards, task assignment and tracking, file-based submissions, and a grading workflow with feedback. The project strengthened practical skills in frontend design, REST API consumption, asynchronous JavaScript, file uploads, and cloud deployment, and demonstrated the ability to independently design and ship a complete, real-world software product within a defined internship timeline.

4.10 Certificates of Completion and Communication Proof

The Certificate of Internship and Certificate of Training issued by Qveto Technologies (Certificate IDs: qveto/5aJ6dW4hx1 and qveto/sPTnr1KE7h respectively), covering the period 07 June 2026 to 08 September 2026, are attached in Chapter 3 of this report as proof of successful completion of the internship program in Web Development.

5. BIBLIOGRAPHY / REFERENCES

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